SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

SYLLABUS
DIGITAL IMAGE PROCESSING

Course Name	Digital Image Processing
Course Code	070123620
Faculty	Engineering
Course Level	3
Course Credit	3
Sub-Committee (Specialization)	Specialization for Curriculum Development

1. Learning Objectives:

The student will be able to:

1. Apply the fundamental concepts of two-dimensional image acquisition, sampling and quantization for digital image representation.
2. Apply image enhancement and spatial-domain filtering techniques to improve image quality.
3. Apply and analyze two-dimensional Fourier and wavelet transform techniques for image processing and multiresolution analysis.
4. Apply morphological operations based on set-theoretic, neighborhood and connectivity concepts for image analysis.
5. Analyze image segmentation and edge-detection techniques for extracting meaningful regions and objects from images.
6. Evaluate and select appropriate image processing techniques for different real-world applications based on their suitability and performance.

2. Pre-requisites

- Signals and Systems
- Digital Signal Processing
- MATLAB/Python basics

॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

3. Course Outline

Unit	Topic	Contents	Hours
1	Digital Image Fundamentals	Digital image representation; elements of Digital Image Processing systems; elements of visual perception; structure of human eye; simple image model; sampling and quantization; basic relationship between pixels; imaging geometry; photographic film.	04
2	Contrast Enhancement, Spatial Filtering and Image Restoration	Histogram; histogram equalization; spatial filtering: averaging, weighted averaging, Gaussian filter, Laplacian, Sobel, Prewitt and order-statistic filters; noise models; noise-removal filters—harmonic, contra-harmonic, order-statistic and adaptive median filters; degradation-function modelling; Wiener filtering; inverse filtering.	10
3	Filtering in Fourier Domain	2-D Discrete Fourier Transform; Fourier-transform properties; low-pass filtering; high-pass filtering; band-pass and band-reject filters; notch filters; homomorphic filtering; Gaussian filters; Butterworth filters.	08
4	Segmentation, Point, Line and Edge Detection	Thresholding; Otsu's method; global thresholds; region splitting and merging; Hough Transform; Watersheds.	07
5	Morphological Image Processing	Neighborhood concepts; adjacency and distance measures; dilation and erosion; opening and closing; region filling; thinning; thickening; skeletons; pruning for binary and grayscale images.	08
6	Wavelets and Multi-resolution Processing	Multiresolution processing; image pyramid; wavelet transforms; Haar transform.	04
7	Image Processing Applications	Applications of transforms in fingerprinting; medical applications such as tumor detection; Magnetic Resonance Imaging analysis using transforms; morphological applications; study of IEEE reference papers covering basic ideas of transforms and their applications.	04
		Total	**45**

॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

4. Course Outcomes

CO	Course Outcome	Bloom's Level
CO1	Explain the basic concepts of two-dimensional signal acquisition, sampling and quantization.	B2 – Explain
CO2	Apply fundamental image manipulation and spatial filtering techniques, including histogram stretching, histogram equalization, contrast enhancement, log transformations and image negatives.	B3 – Apply
CO3	Apply two-dimensional Fourier and wavelet transform techniques and analyze their effects on image filtering and multiresolution processing.	B3-Apply
CO4	Apply morphological operations based on set-theoretic, neighborhood and connectivity concepts for image analysis.	B3 – Apply
CO5	Analyze image segmentation, point, line and edge detection techniques for extracting meaningful regions and objects from images.	B4 – Analyze
CO6	Analyze the scope, issues and applications of image processing systems and evaluate suitable image processing techniques for different applications.	B4/B5 – Analyze/Evaluate

5. CO–PO–PSO Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	–	–	2	–	–	–	–	–	1	–	3	1
CO2	3	2	2	1	3	–	–	–	–	–	1	–	3	2
CO3	3	2	2	2	3	–	–	–	–	–	1	–	3	2
CO4	3	2	2	1	3	–	–	–	–	–	1	–	3	2
CO5	3	3	2	2	3	–	–	–	–	–	1	–	3	1
CO6	3	3	2	2	2	1	-	–	1	–	2	1	3	2

Scale: 3 – Strong correlation; 2 – Moderate correlation; 1 – Low correlation; – – No significant correlation.

॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Constituent of Symbiosis International (Deemed University), Pune

6. CO–Unit Alignment

Unit	Main Content	CO
Unit 1	Digital image fundamentals, sampling, quantization	CO1
Unit 2	Histogram, enhancement, spatial filtering, restoration	CO2
Unit 3	Fourier transform and frequency-domain filtering	CO3
Unit 4	Segmentation, point/line/edge detection	CO5
Unit 5	Morphological image processing	CO4
Unit 6	Wavelets and multiresolution processing	CO3
Unit 7	Image-processing applications	CO6

7. CO–PO–PSO Mapping Justification

CO	Mapping	Justification
CO1	PO1 (3)	Application of engineering, mathematical and signal-processing principles for image acquisition, sampling and quantization.
CO1	PO2 (2)	Analysis of image representation, sampling effects and quantization effects on image quality.
CO1	PO5 (2)	Use of computational tools such as MATLAB/Python for representing, visualizing and analyzing digital images.
CO1	PO11 (1)	Provides foundational knowledge that supports adaptation to evolving digital image-processing techniques and technologies.
CO1	PSO1 (3)	Directly develops fundamental competencies in signal and image processing.
CO1	PSO2 (1)	Provides foundational knowledge for image-based engineering applications.
CO2	PO1 (3)	Engineering knowledge of image enhancement, spatial filtering and restoration.
CO2	PO2 (2)	Analysis of image characteristics and selection of suitable enhancement/filtering techniques.
CO2	PO3 (2)	Application and adaptation of filtering techniques to improve image quality for a

		given problem.
CO2	PO4 (1)	Investigation of the effect of different filters and comparison of performance.
CO2	PO5 (3)	Implementation and analysis using MATLAB/Python and image-processing tools.
CO2	PO11 (1)	Provides foundational knowledge to computational image-processing techniques that supports continued learning of emerging enhancement and filtering methods.
CO2	PSO1 (3)	Develops core image-processing and enhancement skills.
CO2	PSO2 (2)	Supports development of practical image-processing solutions.
CO3	PO1 (3)	Mathematical and engineering knowledge of transforms and frequency-domain image processing.
CO3	PO2 (2)	Analysis of frequency components and interpretation of transform-domain filtering effects.
CO3	PO3 (2)	Application of transform-domain techniques to image-processing problems.
CO3	PO4 (2)	Investigation and analysis of Fourier and wavelet transformations and filters.
CO3	PO5 (3)	Computational implementation using MATLAB/Python and transform-based tools.
CO3	PO11 (1)	Provides exposure to transform-based and multiresolution techniques that supports continued learning of evolving image-processing methods.
CO3	PSO2 (2)	Supports practical image-processing applications.
CO3	PSO1 (3)	Strengthens signal and image-processing competencies.
CO4	PO1 (3)	Mathematical and engineering knowledge of morphology and set theory.
CO4	PO2 (2)	Analysis of object structure, connectivity and neighborhood relationships.
CO4	PO3 (2)	Application of morphological operations to solve image-analysis problems.
CO4	PO4 (1)	Investigation of the effects of morphological operations.
CO4	PO5 (3)	Implement, visualize and analyze using image-processing software/tools.
CO4	PO11 (1)	Provides foundational exposure to morphological image-processing techniques that supports continued learning of evolving image-analysis methods.
CO4	PSO1 (3)	Develops image-analysis and computer-vision-related competencies.
CO4	PSO2 (2)	Supports practical image-processing applications.
CO5	PO1 (3)	Engineering knowledge of segmentation and feature-detection techniques.
CO5	PO2 (3)	Analysis of image regions, boundaries, edges and object characteristics.
CO5	PO3 (2)	Selection and application of segmentation techniques to solve image-analysis problems.

CO5	PO4 (2)	Investigation and comparison of segmentation and detection results.
CO5	PO5 (3)	Implementation and visualization using computational image-processing tools.
CO5	PO11 (1)	Provides exposure to image-segmentation and detection techniques that supports continued learning of evolving image-processing and computer-vision methods.
CO5	PSO1 (3)	Directly supports image-processing and computer-vision competencies.
CO5	PSO2 (1)	Supports application of segmentation techniques to engineering problems.
CO6	PO1 (3)	Comprehensive engineering knowledge of image-processing principles and applications.
CO6	PO2 (3)	Analysis of application requirements and comparison of alternative techniques.
CO6	PO3 (2)	Selection and formulation of suitable image-processing approaches for practical problems.
CO6	PO4 (2)	Investigation, comparison and evaluation of alternative techniques.
CO6	PO5 (2)	Use of modern computational tools to implement and benchmark techniques.
CO6	PO6 (1)	Connection of image-processing techniques with real-world engineering and societal applications.
CO6	PO9 (1)	Development of professional communication through case studies, literature reviews, presentations and documentation of image-processing applications.
CO6	PO7 (1)	Consideration of socially relevant applications such as healthcare and security.
CO6	PO11 (2)	Consideration of practical constraints and performance when selecting an appropriate technique.
CO6	PO12 (1)	Continued learning through current research papers (IEEE) and emerging image-processing techniques.
CO6	PSO1 (3)	Integration of image-processing competencies across multiple applications.
CO6	PSO2 (2)	Ability to select and evaluate practical image-processing solutions.

8. Pedagogy

- Classroom Lectures including flipped classroom
- Online courses
- Mini Projects
- Simulations
- Case studies and literature review of contemporary image-processing applications

9. Evaluation

A. Continuous Assessment

- Unit Tests
- Minor Project
- Assignments/Quiz
- Presentations
- Case Studies

B. End Semester Examination

- Written Examination

10. Benchmarked Courses

Sr. No.	Course	University
1	Digital Image Processing (EC704)	NIT, Trichy
2	Digital Image Processing (ELL715)	IIT, Delhi

11. Recommended Books

Textbooks

- Digital Image Processing — Rafael C. Gonzalez and Richard E. Woods, PHI, 2nd Edition.
- Image Processing, Analysis, and Machine Vision with MindTap — Milan Sonka, Vaclav Hlavac and Roger Boyle, 4th Edition, 2015.
- Digital Image Processing — Pratt, W. K., John Wiley, 2001.

Reference Book

- Fundamentals of Digital Image Processing — A. K. Jain, PHI, 1997.